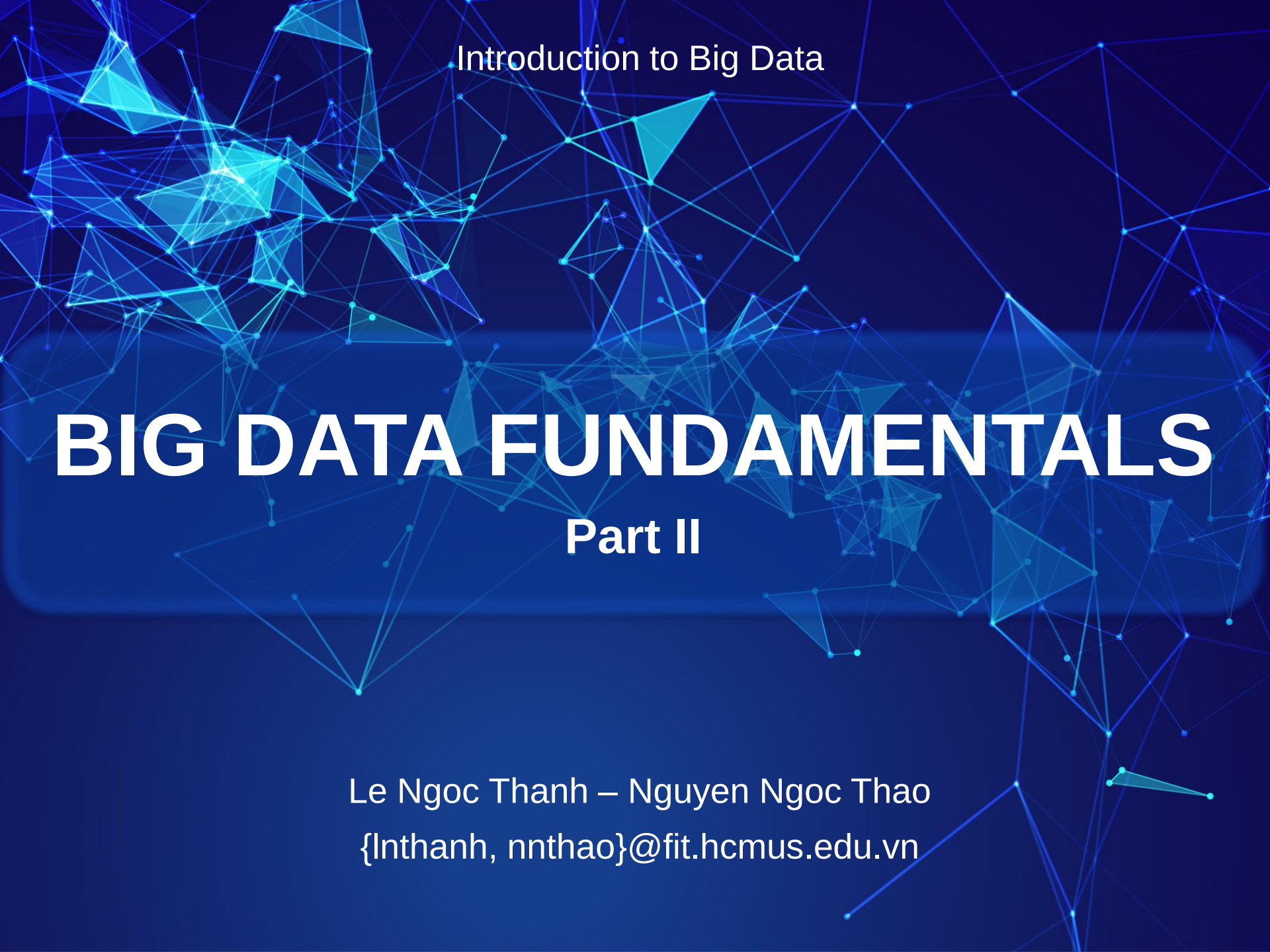




Introduction to Big Data

BIG DATA FUNDAMENTALS

Part II

Le Ngoc Thanh – Nguyen Ngoc Thao
{lnthanh, nnthao}@fit.hcmus.edu.vn

Outline

- Big data platform and Technologies
 - IBM Big data platform
- Digging into Big data technology
 - Big data technology stack
 - Big data analytics platforms and software



Big data platform

Comprehensive,
enterprise-ready,
integrated

Main tasks in Big data

- Tasks within the domain of Big Data often **involve data mining** as a prevalent method, yet at a larger scale.



Data aggregation



Data analysis



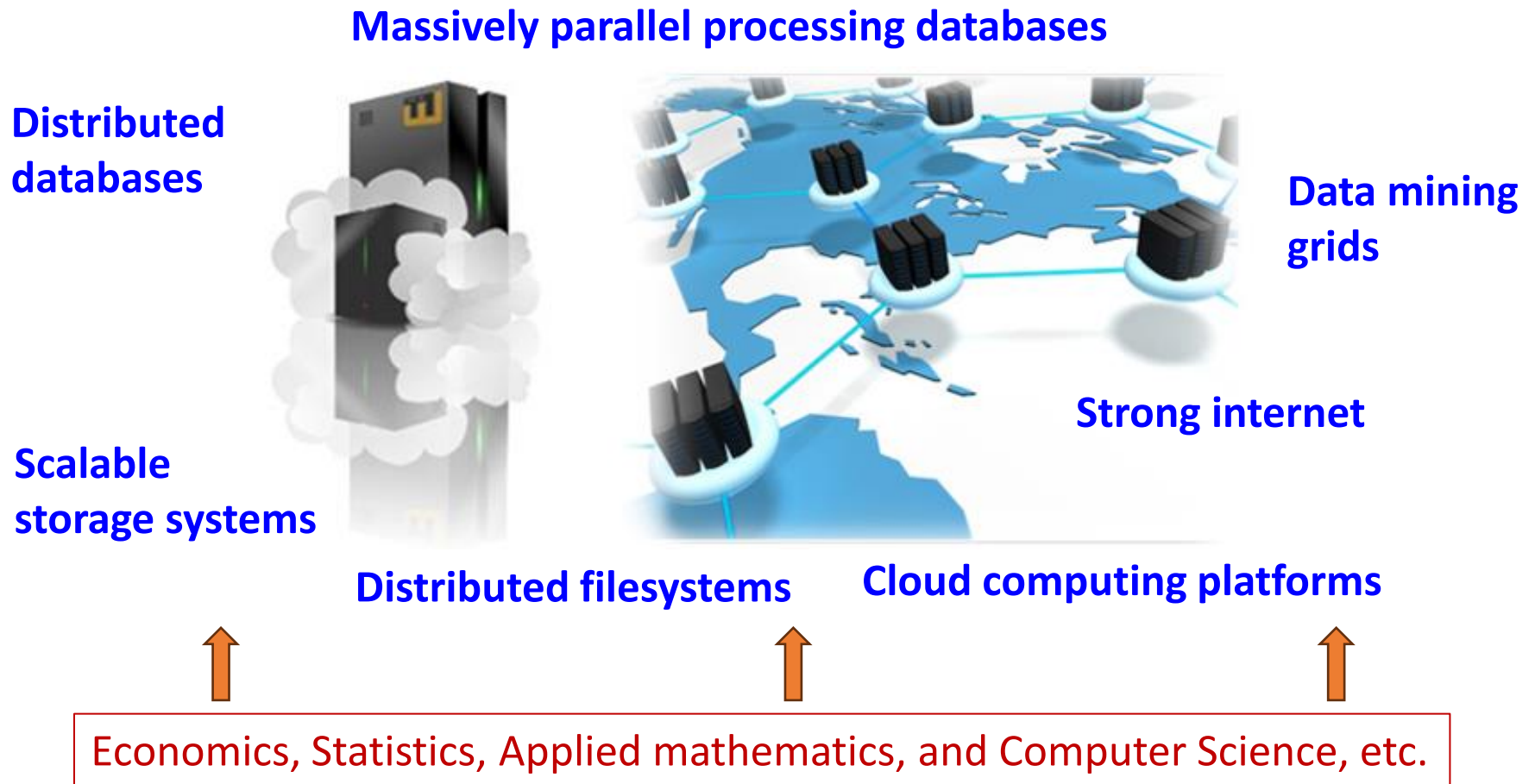
Data
manipulation



Data
visualization

Big data are multidisciplinary

- Technologies in Big data involves multidisciplinary studies.



A Big data platform should offer

Comprehensive

Every **dimension of Big data challenge** is addressed.

Enterprise-ready

Include **features of performance**, security, usability and reliability.

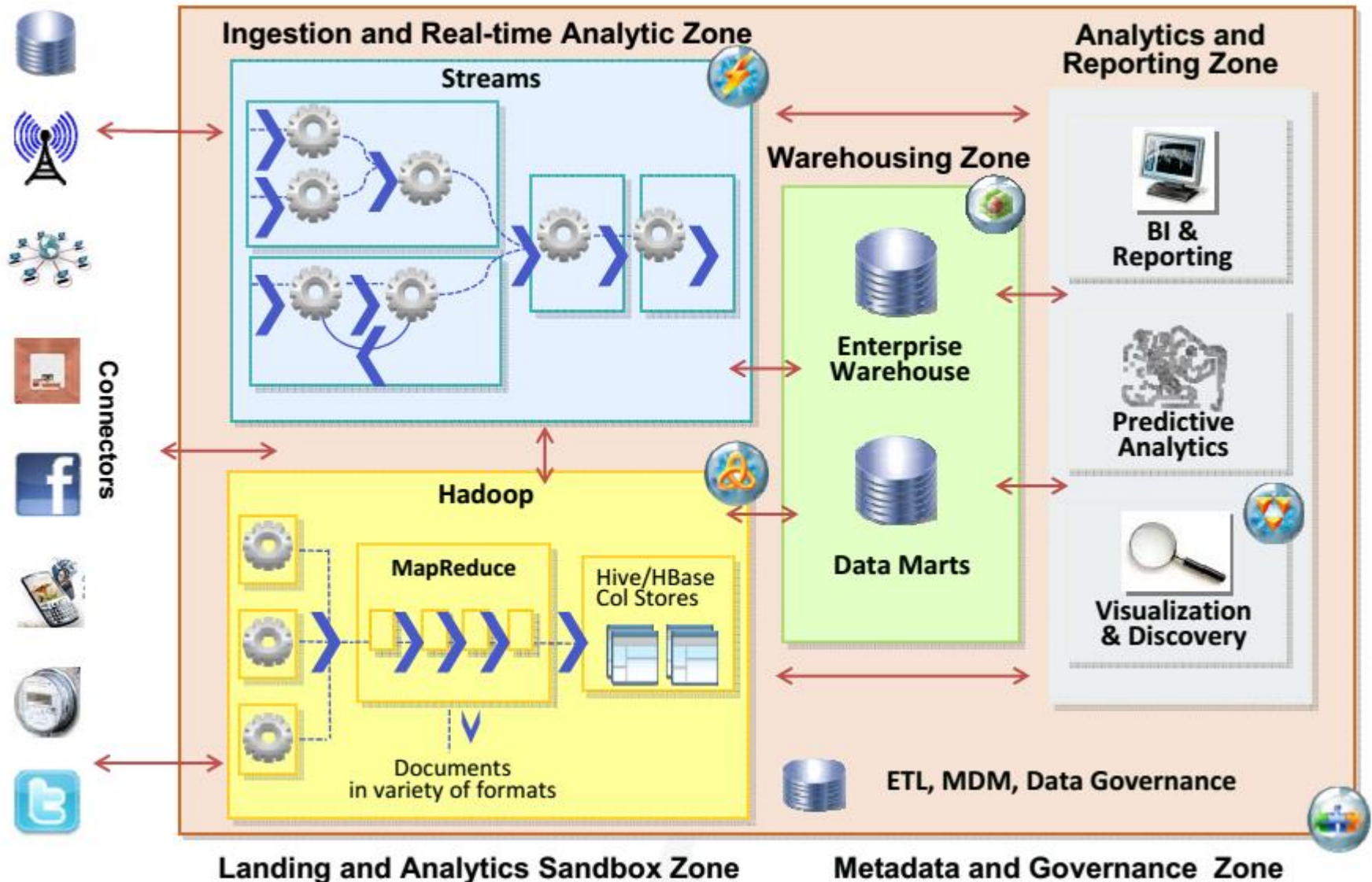
Integrated

Big data technologies to enterprise should be **simplified and accelerated**.

Integration with information supply chain, including databases, data warehouses, and BI applications.

- **Moreover, a Big Data platform should also offer**
 - Open-source based, low latency reads/updates, ad-hoc queries, scalability, extensible, robust fault-tolerant, minimal maintenance.

Components in a Big data platform



IBM Big data platform

- **Strategy:** Move the analytics closer to the data
- Give a solution which is designed specifically with the needs of the enterprise in the mind.
 - Integrate and manage the full variety, velocity and volume of data
 - Apply advanced analytics to information in its native form
 - Visualize all available data for adhoc analysis
 - Development environment for building new analytic applications
 - Workload optimization and scheduling
 - Security and Governance
- Source: [Big Data Platforms, Tools, and Research at IBM](#)

IBM Big data platform

Analytic Applications

BI / Reporting	Exploration / Visualization	Functional App	Industry App	Predictive Analytics	Content Analytics
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IBM Big Data Platform

Visualization
& Discovery

Application
Development

Systems
Management



Accelerators

Hadoop
System



Stream
Computing



Data
Warehouse



Information Integration & Governance



Digging into Big data technology

Digging deeper, better insights

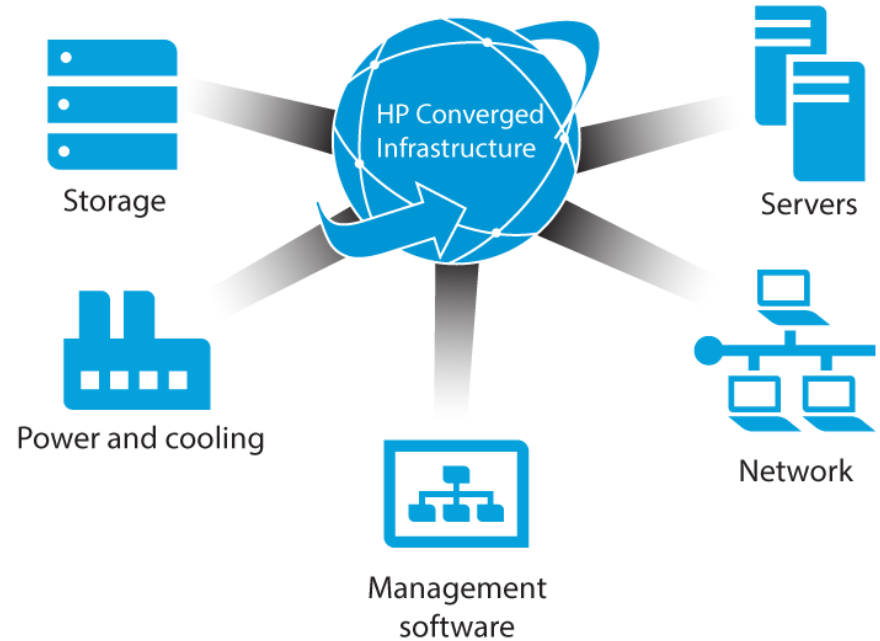


Big data technology stack



Layer 0: Redundant physical infrastructure

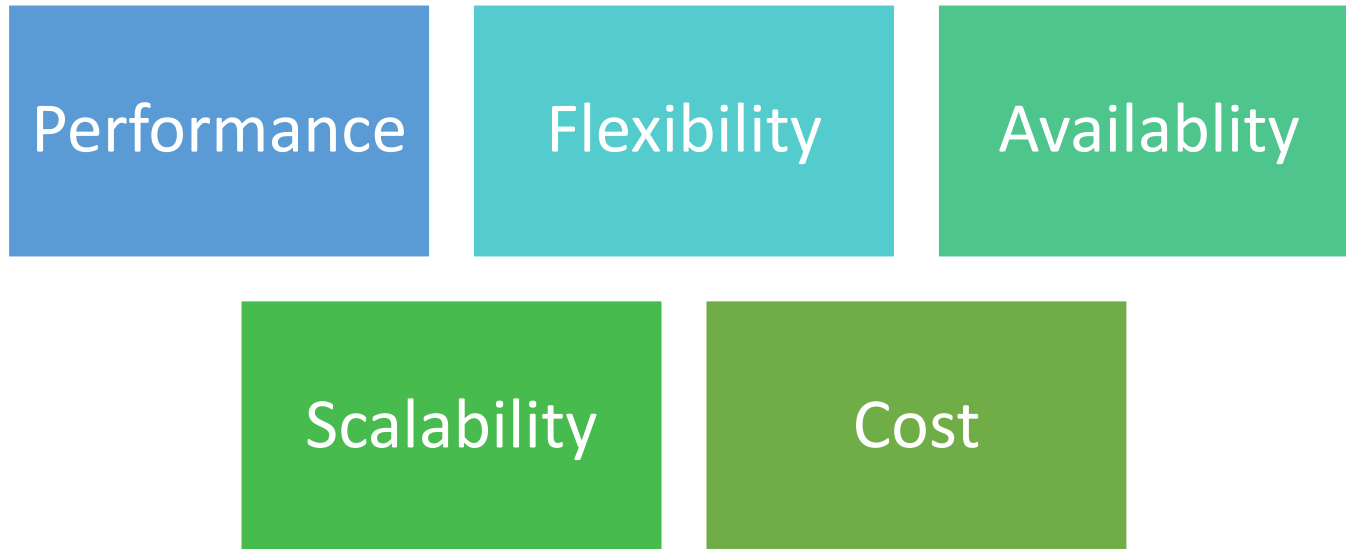
- The **physical infrastructure** is the lowest level.
 - Hardware, network, etc.



- Your company might already have a data center or made investments in physical infrastructures.
- Hence, you may want to find a way to utilize existing assets.

Where most of this began?

- A prioritized list of these principles should include statements about the following



Why redundant?

- Most big data implementations need to be **highly available**.
- That is, networks, servers, and physical storage must be both **resilient** and **redundant**.
- A system is **resilient to failure or changes** when **sufficient redundant resources are in place**, ready to jump into action.



Layer 1: Security infrastructure

- **Security** and **privacy** requirements for big data are similar to those for conventional data environments.
- They must be closely aligned to specific business needs.

Data access

The data should be available only to those who have a legitimate business need for examining or interacting with it.

Protection from unauthorized usage or access are offered by most APIs.

Application access

Data encryption

Most challenging, extremely stress the systems' resources
Encrypt only data elements that require this level of security

The inclusion of mobile devices and social networks exponentially increases both the amount of data and the opportunities for security threats.

Threat detection

Layer 2: Operational databases

- The core of any Big data environment is **database engines** holding collections of data elements relevant to a business.

Atomicity

If any part of the transaction or the underlying system fails, the entire transaction fails.

Only transactions with valid data will be performed.

Consistency

Isolation

Multiple simultaneous transactions do not interfere with each other. All valid transactions will execute until completed and in the order, they were submitted for processing.

After the data from the transaction is written to the database, it stays there “forever.”

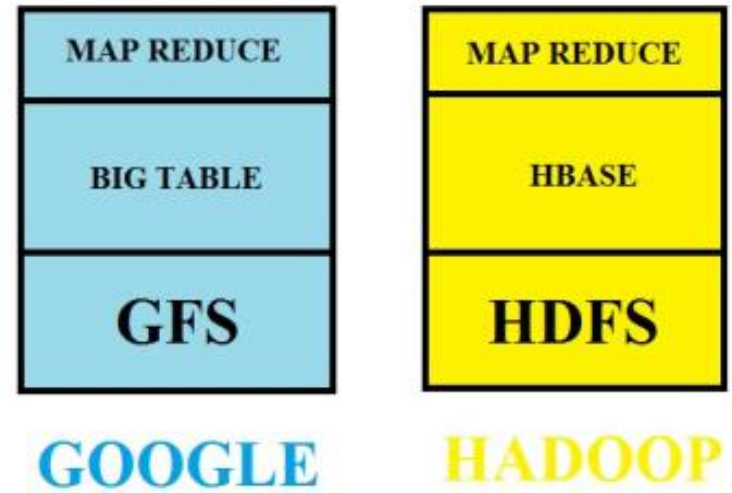
Durability

Layer 3: Organizing Data Services and Tools

- Prepare an ecosystem of tools and technologies to gather and assemble data in preparation for further processing
- Technologies in this layer include the following:
 - A distributed file system
 - Serialization services
 - Coordination services
 - Extract, transform, and load (ETL) tools
 - Workflow services

Hadoop, MapReduce and Big Table

- New technologies to store, access, and analyze huge amounts of data



- Proved to be the sparks that led to a new generation of data management.
- Addressing one of the most fundamental problems: the capability of **processing massive amounts of data efficiently, cost effectively, and in a timely fashion.**

Layer 4: Traditional and advanced analytics

- **What** does your business now **do with all the data** in all its forms to try to make sense of it **for the business**?
 - Managing big data holistically requires different analysis approaches, *depending on the problem being solved*, to help the business to successfully plan.
 - Some analyses will use a traditional data warehouse, while the others will take advantage of advanced predictive analytics.
- **Key techniques:** Analytical data warehouses and data marts, Big data analytics, Reporting and visualization, etc.

Big data platform and analytics software

- Features of Big data platform and analytics software



Data ingestion, Data management, ETL and Warehouse, Hadoop system and Stream Computing

Analytics/Machine learning, Content management, Data integration and governance



Provide efficiency in workplace
Provide accurate data
Give answer to complex questions
It is secure

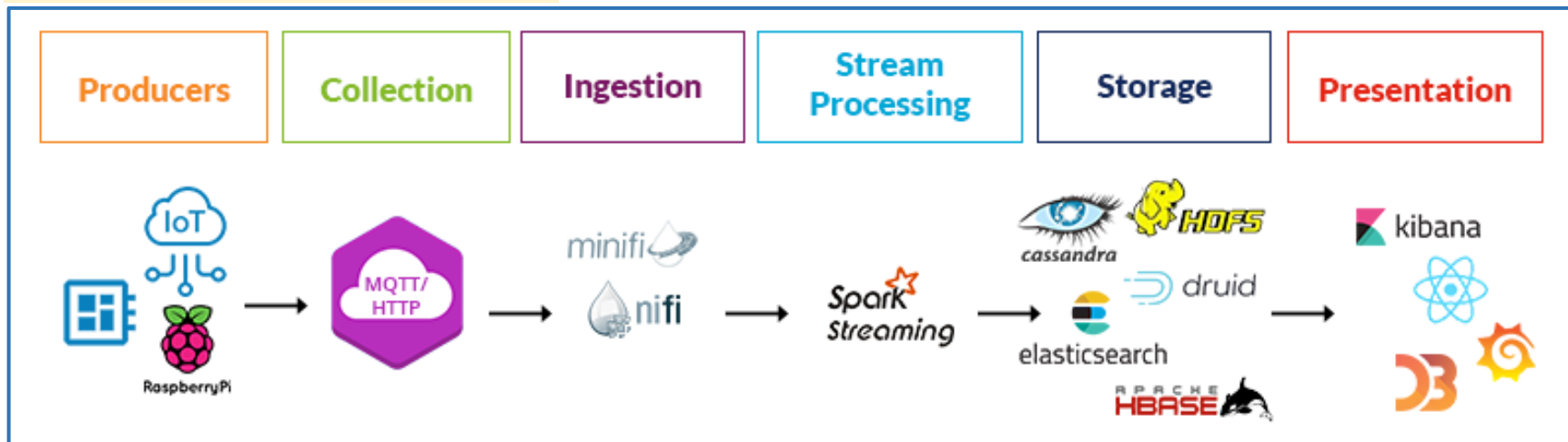
Big data analytic platform tools

- There are some key Big data analytic platform tools available for enterprise use

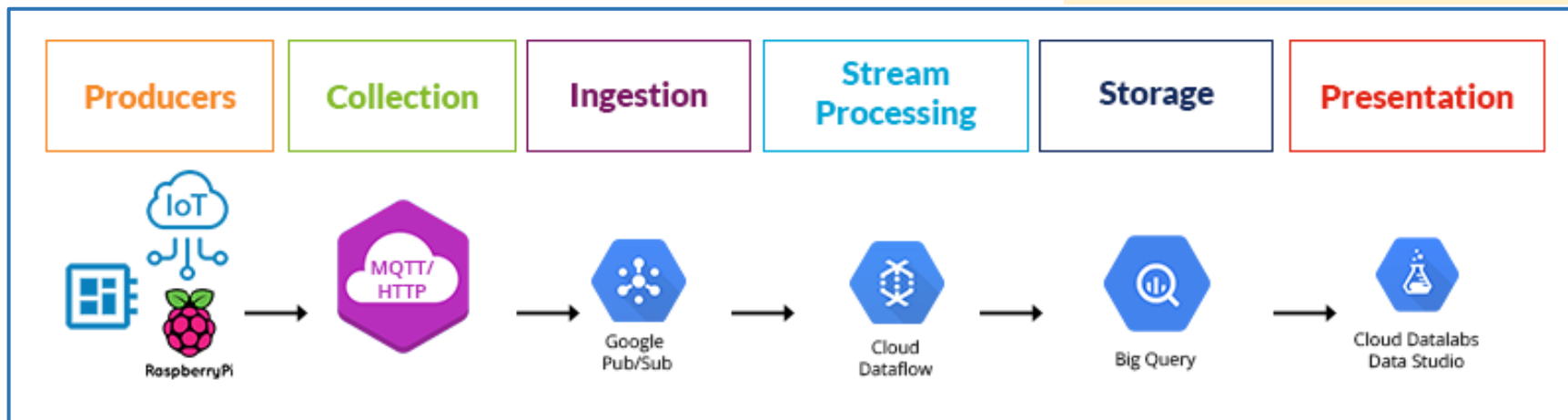


IoT Analytics Platform for Real-Time Data Ingestion

Xenonify IoT Architecture

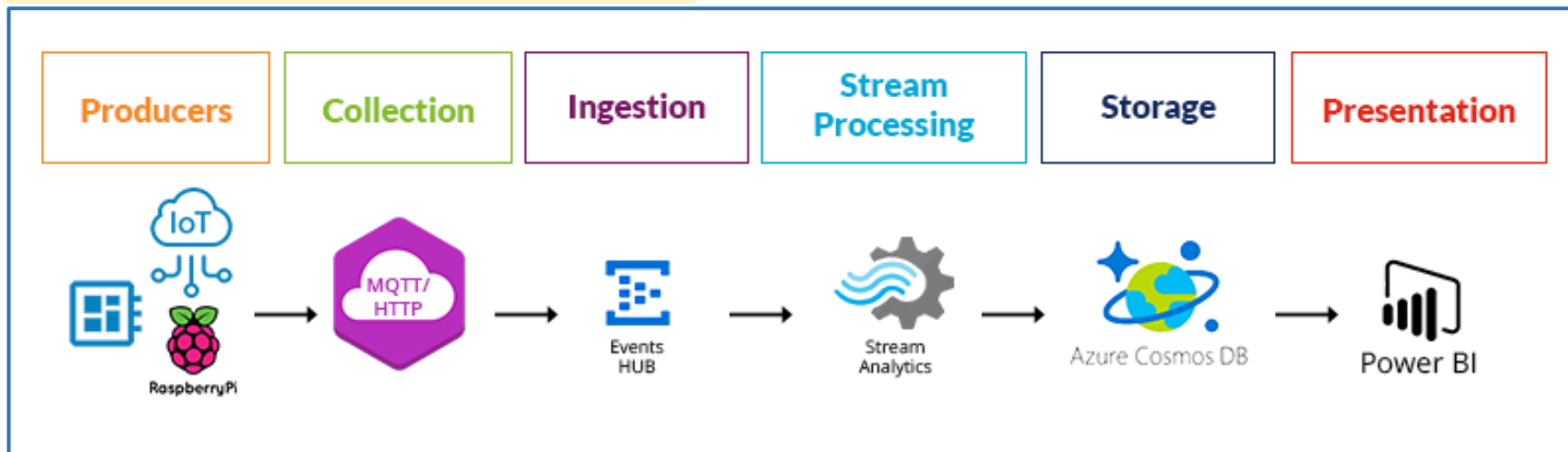


Google IoT Architecture

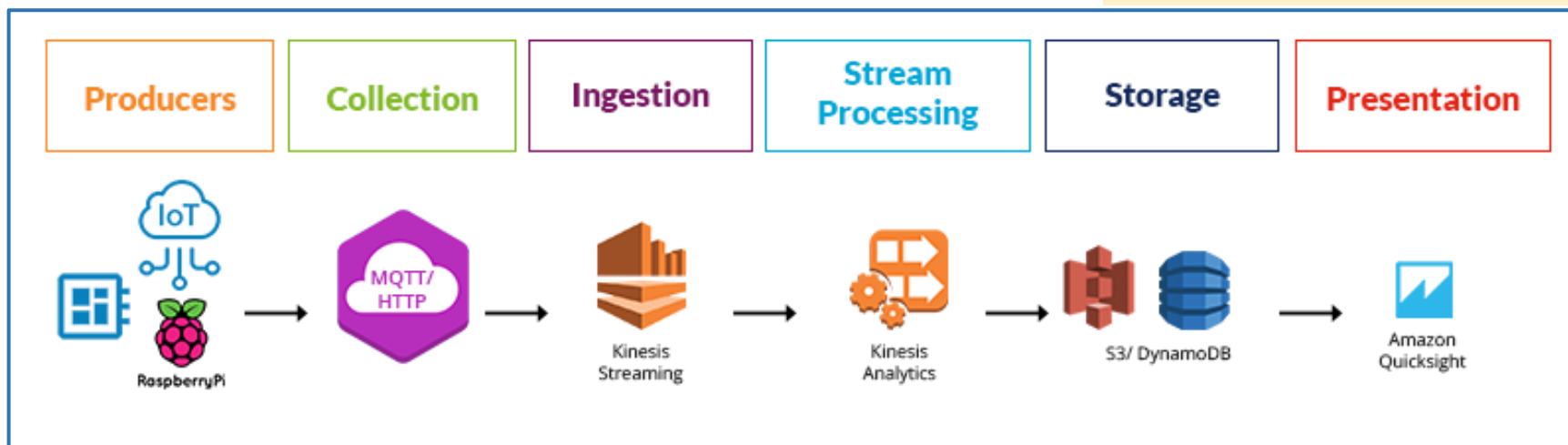


IoT Analytics Platform for Real-Time Data Ingestion

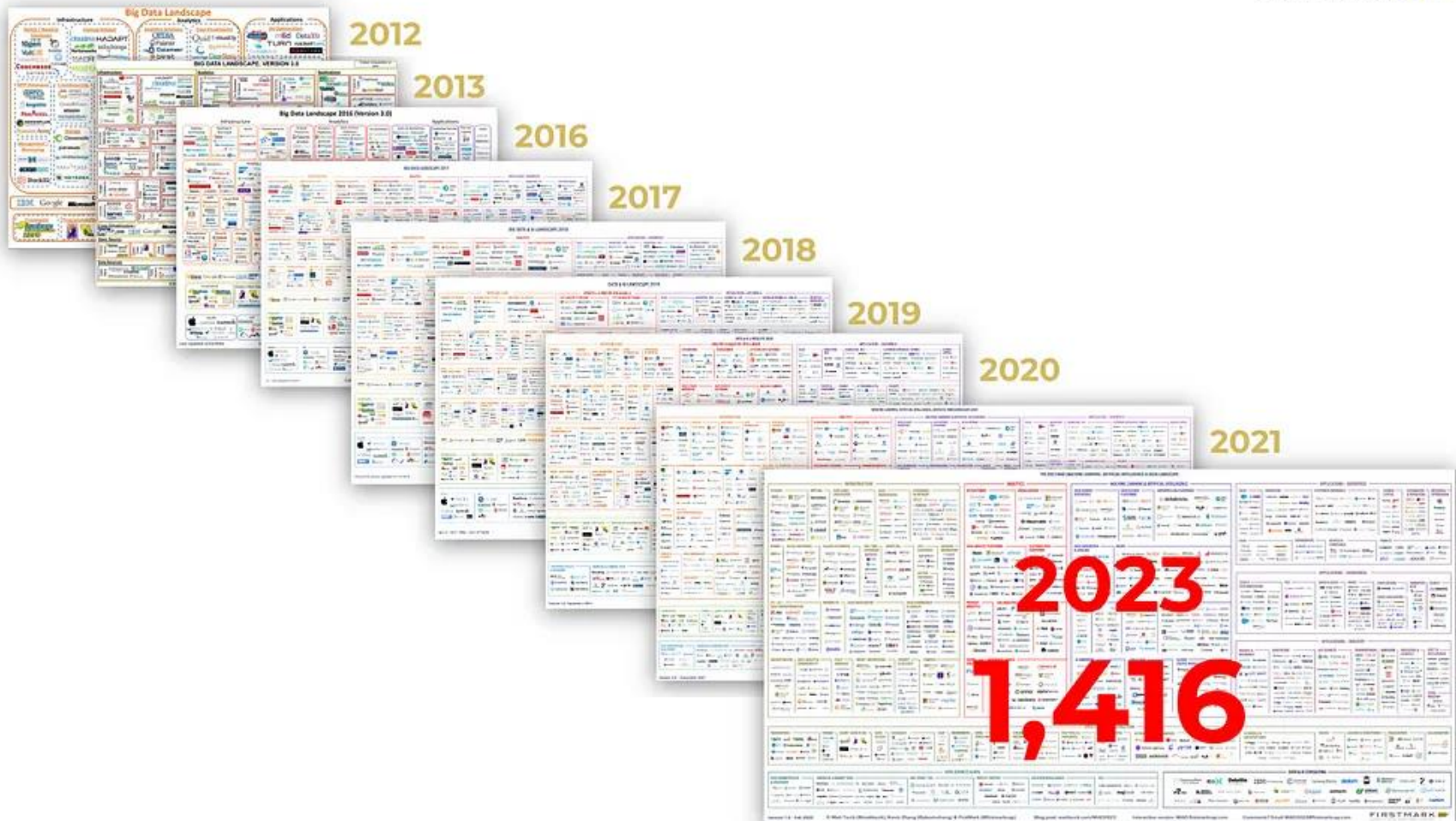
Microsoft Azure IoT Architecture



AWS IoT Architecture



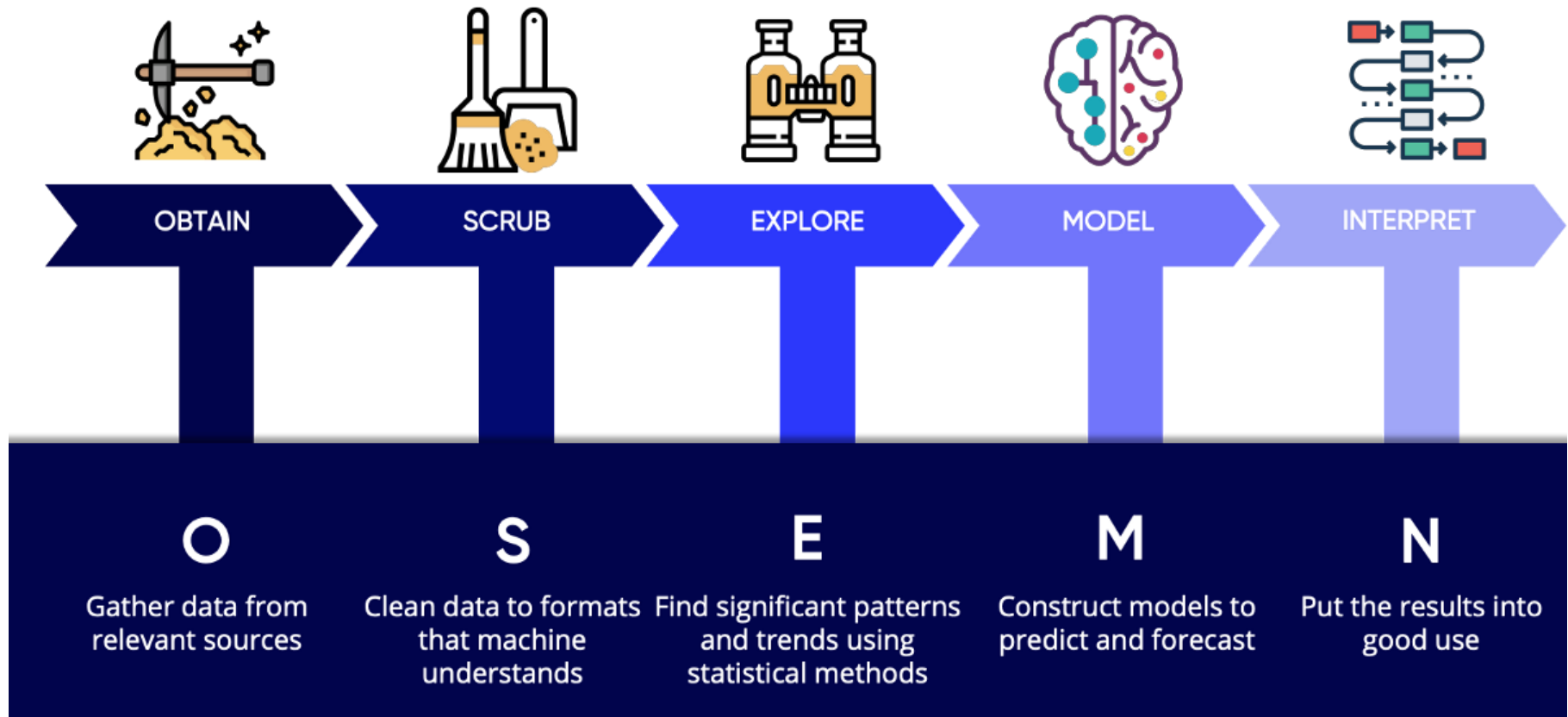
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The 2023 MAD (ML/AI/Data) Landscape

Big data vs. Data science

- In Data science, the data can be of all sizes, which is related to a business or scientific case.
- Big data offers techniques to handle large-scale data at different steps.



...the end.

